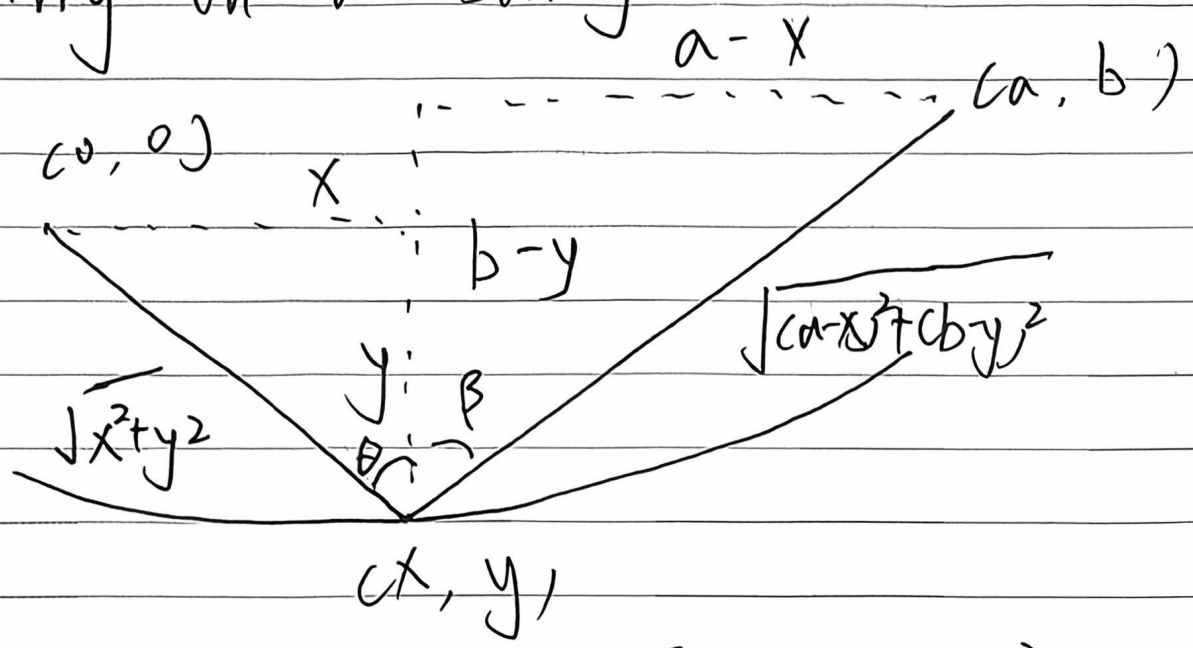
ring on a string




$$\sqrt{x^2+y^2}+\sqrt{(a-x)^2+(b-y)^2}=L \quad (L \text{ is constant})$$

Implicit diff: $y'=0$

$$\frac{1}{2}\left(\frac{2x+2yy'}{\sqrt{x^2+y^2}}\right)-\frac{1}{2}\left(\frac{2(a-x)+2(b-y)y'}{\sqrt{(a-x)^2+(b-y)^2}}\right)=0$$

$$\frac{x}{\sqrt{x^2+y^2}}=\frac{a-x}{\sqrt{(a-x)^2+(b-y)^2}}$$

$$\cancel{\cos\theta=}\ \sin\theta=\frac{x}{\sqrt{x^2+y^2}} \qquad \sin\beta=\frac{a-x}{\sqrt{(a-x)^2+(b-y)^2}}$$

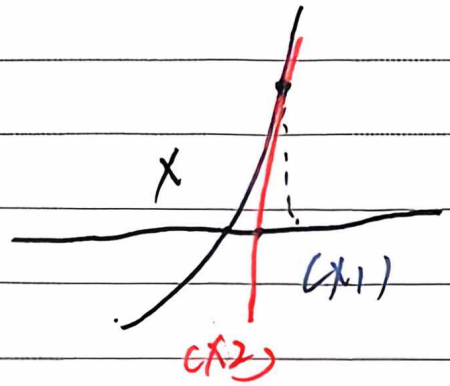
$$\sin\theta=\sin\beta \Rightarrow \theta=\beta$$

# Newton's method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \text{(iterate)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$




$\cancel{E_0 = |x_1 - x_0|}$ $E_0 = |x - x_1|$ $E_1 = |x - x_2|$.

$E_1 \approx E_0^2$.

| $E_0$ | $E_1$ | $E_2$ | $E_3$ | $E_4$ | ..... |
|---|---|---|---|---|---|
| $10^{-1}$ | $10^{-2}$ | $10^{-4}$ | $10^{-8}$ | $10^{-16}$ | ..... |

Newton's method works very well if:

$f'(x)$ not too small

$f''(x)$ not too big.

$x_0$ starts nearby $x$